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MTH 311 — Past Questions & Materials

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Course: MTH 311

Institution: Delta State Polytechnic, Otefe-Oghara

Papers in this pack: 3

Levels: HND1, ND1

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Paper 1 of 3 — ADVANCE ALGEBRA MTH 311 HND1 (20232024) FIRST SEMESTER

Level ND1 • First Semester • Handout

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DELTA STATE POLYTECHNIC, OTEFE-OGHARA
SCHOOL OF APPLIED SCIENCES AND TECHNOLOGY
DEPARTMENT OF STATISTICS

2023/2024 FIRST SEMESTER EXAMINATION.

COURSE CODE/ TITLE: MTH 311/ ADVANCE ALGEBRA

DEPARTMENT: ALL ENGINEERING AND PHYSICS/ELECTRONICS OPTION

LEVEL: HND1

DURATION: 2HRS 30MINS

INSTRUCTION: answer question number one (compulsorily) and any other four (4) questions

1. Taking the scale of 2cm to 1unit on the x axis and 2cm to 2units on the y axis, draw the graphs of $\cosh x$ and $\sinh x$, taking the graphs of e^x and e^{-x} into consideration. On this same graph, draw a linear graph of $x + 3y = 6$. From the graphs, if $y = 3.2$;
 - a. What are the corresponding graphs; x ?
 - b. Find the values of y if the point 2.5cm is drawn perpendicular to intercept $\sinh x$
 - c. Find the values of x on $\cosh x$ with which the graph of $x + 3y = 6$ intercept $\cosh x$
 - d. What is the positive value of x if $y=2.4$ intercepts $\sinh x$?
2. a. Draw the detailed Graphs of
 - (i) $\cosh x$
 - (ii) $\sinh x$ and state the characteristics of each.
- b. Given the series $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots$ which is known to converge, Test the series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$ for convergence, using comparison test
3. a. Let $u = (3, -5, 1)$, $v = (-7, 0, 4)$ and $w = (0, 6, -5)$. Find
 - (i) $4u - 2v$ (ii) $3u + 3v - 5w$ (iii) $\|u\|$ (iv) $\|u-v\|$
- b. If $u = \begin{bmatrix} 4 \\ 1 \\ -2 \\ 3 \end{bmatrix}$ and $v = \begin{bmatrix} 7 \\ -2 \\ 1 \\ 3 \end{bmatrix}$
 - Compute (i) $u+v$ (ii) $2u-3v$ (iii) $\|u\|$ (iv) $d(u,v)$
4. a. Express the series expansion of Hyperbolic functions of
 - (i) $\sinh x$ (ii) $\cosh x$ using the series power expansion of exponential function e^x and e^{-x} respectively.
- b. Find the scalar product of $\vec{A} = 4i - 3j + 5k$, $\vec{B} = 3i + 2j - 2k$.
5. a. find the interval of convergence of the series;
 - (i) $\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ (ii) $\sum_{n=0}^{\infty} \frac{2^n x^n}{n!}$
- b. Find the Maclaurin series expansion of;
 - (i) $\ln(1+x)$ (ii) $\cos x$
6. a. (i) Using L'Hopital's Rule, evaluate the limiting values of

$$\lim_{x \rightarrow -2} \frac{16x^3 + 6x^2 + 3x + 4}{8x^3 + 4x^2 + 2x + 2}$$
- b. write out the matrix of the quadratic form of $3x_1^2 + 4x_1x_2 - 2x_2^2 + 2x_1x_3 + 2x_3^2$, and show that the said matrix of the quadratic form is positive definite.

2. a. Draw the detailed Graphs of
 (i) $\cosh x$
 (ii) $\sinh x$ and state the characteristics of each.
- b. Given the series $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots$ which is known to converge, Test the series $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots$ for convergence, using comparison test
3. a. Let $u = (3, -5, 1)$, $v = (-7, 0, 4)$ and $w = (0, 6, -5)$. Find
 (i) $4u - 2v$ (ii) $3u + 3v - 5w$ (iii) $\|u\|$ (iv) $\|u - v\|$
- b. If $u = \begin{bmatrix} 4 \\ 1 \\ -2 \\ 3 \end{bmatrix}$ and $v = \begin{bmatrix} 7 \\ -2 \\ 1 \\ 3 \end{bmatrix}$
 Compute (i) $u + v$ (ii) $2u - 3v$ (iii) $\|u\|$ (iv) $d(u, v)$
4. a. Express the series expansion of Hyperbolic functions of
 (i) $\sinh x$ (ii) $\cosh x$ using the series power expansion of exponential function e^x and e^{-x} respectively.
- b. Find the scalar product of $\vec{A} = 4i - 3j + 5k$, $\vec{B} = 3i + 2j - 2k$.
5. a. find the interval of convergence of the series;
 (i) $\sum_{n=1}^{\infty} \frac{x^n}{n3^n}$ (ii) $\sum_{n=0}^{\infty} \frac{2^n x^n}{n!}$
- b. Find the Maclaurin series expansion of;
 (i) $\ln(1+x)$ (ii) $\cos x$
6. a. (i) Using L'Hopital's Rule, evaluate the limiting values of
 Limit $\frac{16x^3 + 6x^2 + 3x + 4}{8x^3 + 4x^2 + 2x + 2}$
- b. write out the matrix of the quadratic form of $3x_1^2 + 4x_1x_2 - 2x_2^2 + 2x_1x_3 + 2x_3^2$, and show that the said matrix of the quadratic form is positive definite.
7. a. Given the 2×2 matrix; $A = \begin{bmatrix} 2 & 3 \\ 4 & 1 \end{bmatrix}$
 (i) find the characteristic matrix
 (ii) find the characteristic polynomial
 (iii) The characteristic equation
 (iv) The Eigen values
- b. Find the corresponding Eigen vectors of matrix A
- DR. OGBOGBO G. O
(EXAMINER)

Paper 2 of 3 — MTH 311 HND1 (20222023) FIRST SEMESTER ADVANCE ALGEBRA

Level HND1 • First Semester • Handout

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DELTA STATE POLYTECHNIC, OTEFE-OGHARA
SCHOOL OF APPLIED SCIENCES AND TECHNOLOGY
DEPARTMENT OF STATISTICS
2022/2023 FIRST SEMESTER EXAMINATION.

COURSE CODE: MATH 311

COURSE TITLE: ADVANCE ALGEBRA

DEPARTMENT: ALL ENGINEERING

LEVEL: HNDI

DURATION: 2HRS 30MINS

INSTRUCTION: ANSWER QUESTION ONE (1) AND ANY OTHER FOUR (4) QUESTIONS

1. Taking the scales of 2cm to 1 unit on x axis and 3cm to 2 units on y axis, draw the graphs of $\cosh x$ and $\sinh x$. Taking the graphs of e^x and e^{-x} into consideration. On this same graph, draw the linear graph of $2x + 3y = 6$. From the graphs, if $y=3.2$
 - (i) What are the corresponding graphs; x ?
 - (ii) Find the values of y if the point 2.8cm is drawn perpendicular to intercept $\sinh x$
 - (iii) Find the value of x on $\cosh x$ with which the graph of $2x + 3y = 6$ intercept $\cosh x$.
 - (iv) What is the positive value of x if $y=2.7$ intercepts $\sinh x$
2. a. Find the range of values of x , for which the following series is convergent or divergent
 - i. $\sum \frac{(n+1)x^n}{n^3}$
 - ii. $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.5} + \dots$
- 2.b. Find the Maclaurin series expansion of
 - (i) $\ln(1+x)$ (ii) $\cos x$
- 3.a. The 3rd and 7th terms of the G.P are 81 and 16 respectively. Find the series.
- b. Given $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ as infinite G.P. find the limit δ_n and hence deduce whether the above infinite series convergent or not
4. a. Let $u = (2, -7, 1)$, $v = (-3, 0, 4)$ and $w = (0, 5, -8)$. Find
 - (i) $3u - 4v$ (ii) $2u + 3v - 5w$ (iii) $||u||$ (iv) $||u-v||$
- b. If $u = \begin{bmatrix} 7 \\ 1 \\ -3 \\ 6 \end{bmatrix}$ and $V = \begin{bmatrix} 3 \\ -2 \\ 1 \\ 4 \end{bmatrix}$
 - Compute (i) $u+v$ (ii) $2u-3v$ (iii) $||u||$ (iv) $d(u,v)$
- 5.a. Express the series expansion of Hyperbolic functions of
 - (i) $\sinh x$ (ii) $\cosh x$, using the series power expansion of exponential function e^x and e^{-x} respectively.
- 5.b. Find the scalar product of $\vec{A} = 3i - 2j + 5k$, $\vec{B} = 2i + 3j - 2k$.
- 6.a. Test the following series for convergence
 - (i) $\sum \frac{n}{2^n}$ (ii) $\sum \frac{(-1)^n 3^n}{n^2}$
- b. Write out the matrix of the quadratic form $2x_1^2 - x_1x_2 + x_2^2 + 4x_1x_3 - 3x_3^2$ and determine whether it is positive definite.
- 7.a. (i) Using L'Hopital's Rule, evaluate the limiting values of

$$\text{Limit } \frac{8x^3 + 5x^2 + 2x + 1}{4x^3 + 6x^2 + 7x + 3}$$
- b. (i) find the log form of $\tanh^{-1} x$ (ii) Evaluate from (bi) above the value of $\tanh^{-1} 0.5$

EXAMINER

DR. OGBORO GABRIEL O.

Paper 3 of 3 — MTH 311 HND1 (2021/2022) FIRST SEMESTER ADVANCE ALGEBRA

Level HND1 • First Semester • Handout

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DELTA STATE POLYTECHNIC, OTEFE-OGHARA
SCHOOL OF APPLIED SCIENCES & TECHNOLOGY
DEPARTMENT OF STATISTICS

FIRST SEMESTER 2021/2022 ACADEMIC SESSION

COURSE CODE: MTH 311
COURSE TITLE: ADVANCE ALGEBRA
DEPARTMENT: HNDI, ALL ENGINEERING
INSTRUCTION: ANSWER ANY FIVE QUESTIONS
TIME: 2½ HOURS

QUESTION ONE

1a. Determine whether each of the following series is convergent or divergent

(i) $\sum \frac{n+1}{n^2+2}$ (ii) $\sum \frac{1+2n^2}{1+n^2}$

b. (i) Find the limiting value of $\lim_{n \rightarrow \infty} \frac{n^2 - 3n + 4}{4n^2 + 5n + 1}$

(ii) If $\sum_{n=1}^{\infty} a_n$ is a series of positive term, and $\sum_{n=1}^{\infty} b_n$ is a series of positive

terms that is known to converge, Test the series $1 + \frac{1^2}{2} + \frac{1^3}{3} + \frac{1^4}{4} + \dots$

Comparing the series $1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$ that is known to converge.

QUESTION TWO

2a. Find the interval of convergence of

(i) $\sum_{n=0}^{\infty} \frac{2^n x^n}{n!}$ (ii) Find the range of values of x for which the following series is absolutely convergent. $\sum n^n x^n$

b. Find the Maclaurin series expansion of $\sin x$.

QUESTION THREE

3. Define the following with examples

- (i) Null matrix
- (ii) Transpose matrix
- (iii) Symmetric and skew-symmetric matrix
- (iv) Diagonal matrix
- (v) Unit matrix

QUESTION FOUR

- 4a. Draw the common Graphs of $\sinh x$ $\cosh x$ and $\tanh x$ and state their characteristics.

- b. Find the inverse of the matrix $A = \begin{bmatrix} 3 & 2 & 2 \\ 1 & 3 & 2 \\ 1 & 4 & 1 \end{bmatrix}$.

QUESTION FIVE

- 5a. The 3rd and 7th terms of a G.P are 81 and 16 respectively
Find the series

- b. Test $\sum \frac{(-1)^n 3^n}{2^n}$ for convergence

QUESTION SIX

- 6a. Solve the equation using Cramer's Rule

$$\begin{aligned} 2x - y + 3z &= 2 \\ x - 3y - z &= 11 \\ 2x - 2y + 5z &= 3 \end{aligned}$$

- b. Find the eigen values of the matrix $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$

QUESTION SEVEN

- 7a. Define stokes theorem

- b. Verify stokes theorem for $\vec{F} = -y\vec{i} + 2yz\vec{j} + y^2\vec{k}$ where S is the upper half of the sphere $x^2 + y^2 + z^2 = 4$

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EXAMINERS